

Design review #3 – Ring oscillator



Project developed by:	SLICE Semiconductor
Project funded by:	nl-net foundation
Document ID:	DR3: Ring oscillator
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Date created:	22/08/2026
Date modified:	
Revision:	A

Table of Contents

1	General information	2
1.1	Project goal	2
1.2	Specifications	2
1.3	Purpose of Design review #3 (DR3)	3
1.4	Repository details	3
1.5	Open source tools	4
1.6	Open source PDK	4
2	Top-Level design description	4
2.1	Architecture	4
3	Testbench description	6
3.1	Testbench details	6
4	Design hierarchy	6
5	Layout review	7
6	Testbench simulation results	9
6.1	Functionality	9
6.2	Frequency trim	10
6.3	Resolution	11
6.4	Linearity	12
6.5	Frequency spread	13
6.6	Duty cycle and power consumption	14
7	Conclusion	14
8	References	15
9	Revision History	15

1 General information

The primary aim of this project is to develop a high accuracy, low noise, space graded INA. In parallel with that, a 6b tunable ring oscillator (RO) has been developed to provide the following support:

- Simple verification of the I2C / SPI by sweeping codes and observing the change in oscillation frequency (f_{osc}).
- Simple correlation between silicon and simulation (i.e. iHP device models) by comparing f_{osc} over all codes pre and post silicon.
- Physical verification flow (LVS / DRC / PEX) for this project was pipe cleaned and verified using this simple (quick to simulate) block.
- Transition from iHP's SG13G2 → SG13CMOS5L process models was pipe cleaned and verified using this simple (quick to simulate) block.
- Since the RO output is brought off chip, it provides a nice demonstration of the project's capabilities and interaction of the analog and digital blocks together on chip – user can see an oscillating waveform on an oscilloscope whose f_{osc} varies with the application of a trim code.
- An RO can be used as a crude temperature sensor. As accurate temperature sensors will be placed on the ASIC, the performance of an “RO based temperature sensor” can be accurately characterized to lay the foundations for a future possible project.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [1]).

1.2 Specifications

Below specifications to be met across full process (SS/FF/SF/FS), voltage ($\pm 5\%$) and temperature ($-40 \rightarrow 125^\circ\text{C}$) (i.e. PVT) conditions.

Parameter	Unit	Value	Comment
vdda_lv	V	1.5	Digital supply
vssa	V	0	Ground
idda_hv	mA	1	Digital supply current
Power	mW	1.5	Digital power consumption
Fosc (typical conditions)	MHz	10	TT/27C/mid-code
Resolution	%	1	Accuracy
Fosc spread	$\pm$ %	5	Post trim spread
Duty cycle max	%	55	
Duty cycle min	%	45	
Dfosc	$\pm$ %	20	Trim range

Table 1: RO specifications

1.3 Purpose of Design review #3 (DR3)

Purpose of DR3 is to review final layouts before delivery to top level (i.e. chip level) integration. Therefore, upon approval of DR3, layouts are “frozen” and top level integration commences. Specifically, this DR3 refers to the layout freeze of the INA top level (layout freezes for blocks outside of the INA top level are detailed in separate DR3’s).

Note: DR2 is checked in at [2].

1.4 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

As this document pertains to DR3, all material relating to it is found in the sub directory “Design_Review_3” (folder = RO).

As block schematics relate to all DR’s, these are contained in the separate folder “Schematics_Layout”.

Inside each folder here, lives the following views:

- Schematic (to be frozen at DR2)
- Symbol (to be frozen at DR2)
- Layout (to be frozen at DR3)

Checked in schematic / symbol / layout for the top level is shown in Figure 1.

Space_grade_Instrumentation_Amplifier_ASIC / Schematics_Layout / sunrise_ro_core_wcdac_buf / 

 DiarmuidSLICE adding final tapeout schematics

Name	Last commit message
 ..	
 Readme.txt	Create Readme.txt
 sunrise_ro_core_wcdac_buf.gds	adding final tapeout schematics
 sunrise_ro_core_wcdac_buf.sch	Update schematics and symbols.
 sunrise_ro_core_wcdac_buf.sym	Add files via upload

Figure 1 Checked in schematics for ring oscillator top level

1.5 Open source tools

The open source tools used for DR3 are summarized in Table 2, with links to install each tool provided as references.

Tool	Description
ngspice	Spice simulator [3]
xschem	Schematic / waveform viewer [4]
openVAF	verilogA compiler [5]
Octave	Post processing [6]
klayout	Layout viewer / editor [7]
magic	Pcell generator / PEX generation [8]
netgen	LVS [9]

Table 2 Open source tools used for DR3

1.6 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [10]. This is a derivative of their more general option – SG13G2 [11] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm² as opposed to €2.8k/mm² for SG13G2) and runs through the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [12].

2 Top-Level design description

2.1 Architecture

Any ring oscillator consists of 'N' inverting stages where $N \geq 3$, as shown in Figure 2.

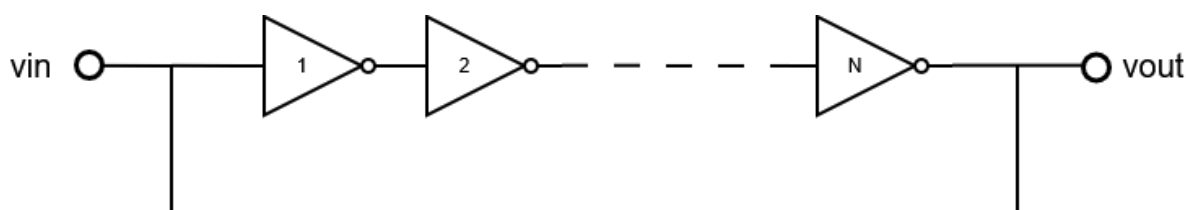


Figure 2 Basic ring oscillator architecture

The resulting oscillation frequency is defined as:

$$f_{osc} = \frac{1}{2t_{pd}N} \dots (1)$$

where:

- t_{pd} = propagation delay through an inversion stage

As t_{pd} is the time it takes the output of an inversion stage to reach vdd/2, it can also be expressed as:

$$t_{pd} = 0.693\tau = 0.693R_{op}C_{in} \dots (2)$$

where:

- R_{op} = effective output resistance of the inverter
- C_{in} = input capacitance of the inverter

Note: in (2), R_{op} varies as a function of the input signal swing due to changes in v_{gs}

Using typical values of $C_{in} = 20fF$, $R_{op} = 5k\Omega$, for $N = 5 \Rightarrow f_{osc} \approx 3GHz$. To get down to the 10MHz spec would require $N \approx 1000$ which is an impractically large number that would result in enormous power consumption. A more efficient approach to reducing the frequency of a ring oscillator is to insert an RC network at each stage, as shown in Figure 3.

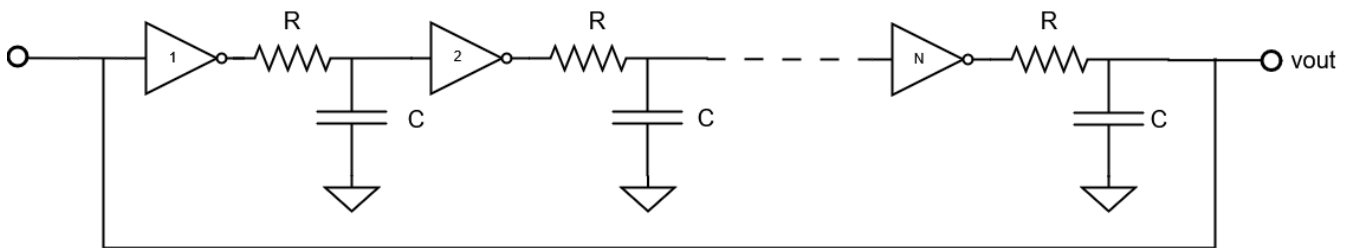


Figure 3 RC based ring oscillator

By sizing $R \gg R_{op}$ and $C \gg C_{in}$, using (1) and (2), f_{osc} can be approximated as:

$$f_{osc} = \frac{1}{2 * 0.693 * RC * N} \dots (3)$$

RC can then simply be sized accordingly for a given frequency.

Note: An interesting observation is that Δf_{osc} in Figure 3 is relatively independent on vdd while Δf_{osc} in Figure 2 varies massively with it (MHz/mV sensitivity). This gives the architecture of Figure 3 far superior power supply rejection (PSR) than that of Figure 2.

The output frequency of any ring oscillator (RO) will vary across PVT conditions where this spread in f_{osc} (Δf_{osc}) is fundamentally different between Figure 2 and Figure 3.

In Figure 2, Δf_{osc} is dominated by ΔR_{op} and so becomes a function of Δv_{th} , v_{dd} , and temperature (T). The minimum f_{osc} (f_{min}) thus occurs at SS, LT, $v_{dd_{min}}$ corners with f_{max} occurring at FF, HT, $v_{dd_{max}}$ corners, where typical spreads are $\approx \pm 30\%$.

In Figure 3, Δf_{osc} is purely a function of ΔR (and hence T) and ΔC . f_{min} thus occurs at R_{max} , C_{max} corners while f_{max} thus occurs at R_{min} , C_{min} corners, where typical spreads are $\approx \pm 20\%$.

To account for these spreads and achieve the target f_{osc} across PVT conditions, a frequency tuning mechanism (or frequency trim) must be included in the architecture whose trim range spans the expected Δf_{osc} range.

Two methods exist for tuning a ring oscillator (RO). The first is by current starving whereby the current into each inverter stage is adjusted to give a linear change in frequency. The second is by placing a CDAC on the RO output as shown in Figure 4.

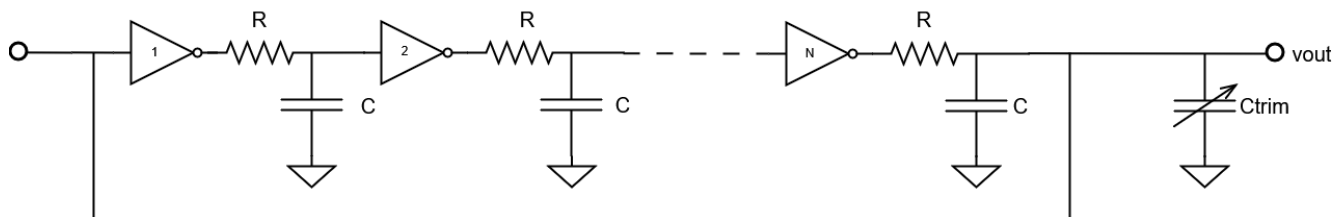


Figure 4 Trimmable RC based ring oscillator

With Ctrim present, f_{osc} is described as:

$$f_{osc} = \frac{1}{C_{trim} + (2 * 0.693 * RC * N)} \dots (4)$$

Due to its simplicity, the CDAC based approach was chosen using a 6b CDAC.

3 Testbench description

3.1 Testbench details

To verify all layouts, a 1:1 comparison of simulation results is made between the layout for the block under test and its corresponding schematic, in the same testbench. This is achieved by simply replacing the schematic view of the block under test, with a parasitic extracted (PEX) netlist describing the corresponding layout.

This means the testbenches detailed in section 3.1 of DR2 (to verify schematics) are re-used in DR3, to verify their corresponding layouts. Reader is thus referred to DR2 [2] for testbench details and how to run them.

4 Design hierarchy

For ease of layout and physical verification (PV) checks, design has been constructed into the multi-layer hierarchy summarized in Table 3.

Level	1	2	3	4
	sunrise_ro_core_wcdac_buf	sunrise_ro_core_wcdac	sunrise_ro_core	sunrise_ro_rc
				sg13cmos5l_nand2
				sg13cmos5l_inv8
				sg13cmos5l_inv16
		sg13cmos5l_inv4 sg13cmos5l_inv16	sunrise_ro_cdac	sunrise_ro_cdac_cap_array

Table 3 Design hierarchy

RO top level symbol is shown in Figure 5.

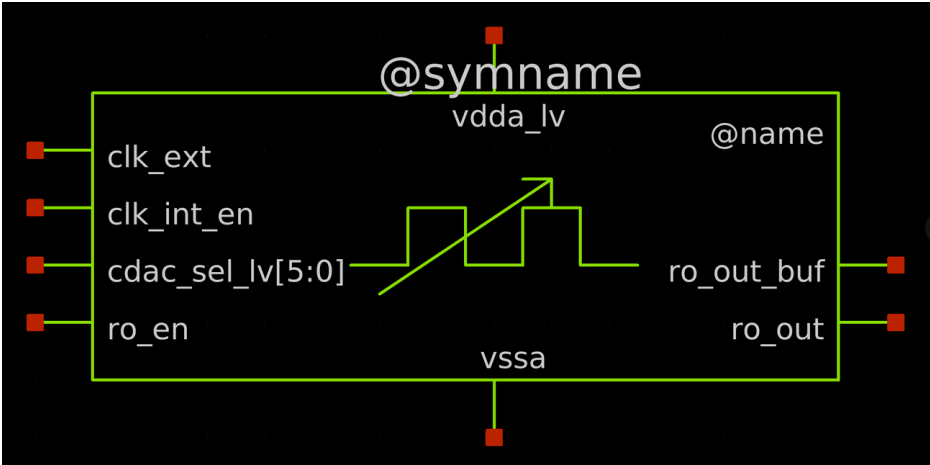


Figure 5 RO top level symbol

One change from DR2 is the addition of a MUX at the output. This is for risk mitigation in the unlikely event the oscillator will not start up in silicon. As the I2C/SPI on the ASIC requires a clock, in such an event, an external clock gets MUX'ed through instead of the internal one. Schematic of this MUX is shown in the orange box of Figure 6 with the external clock input shown in the red box.

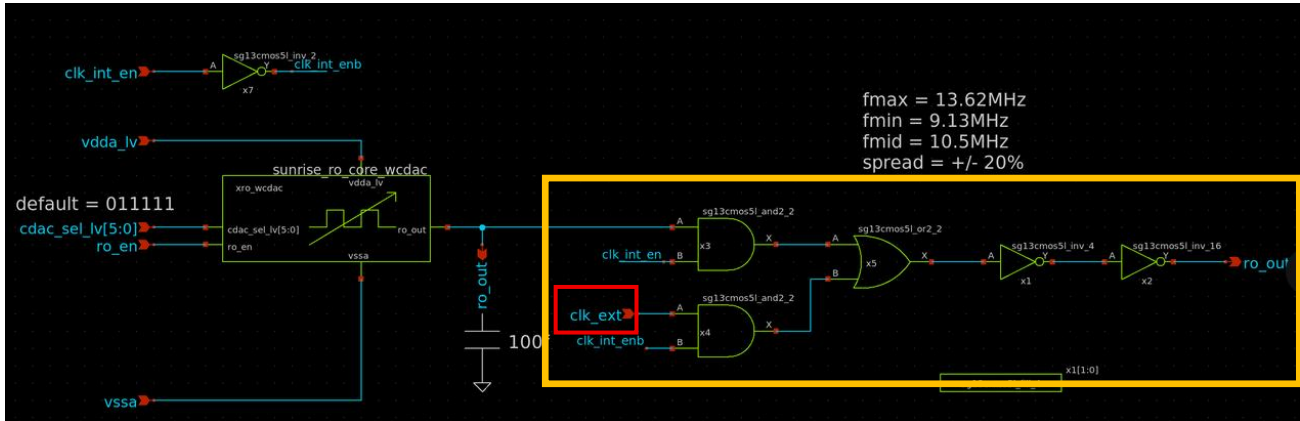


Figure 6 Newly added output MUX post DR2

5 Layout review

Top level layout is shown in Figure 7 **Error! Reference source not found.** where it can be seen to occupy a footprint of ~ 130 x 75um.

Top level layout is delivered with a full power grid on M4/M5. Figure 8 is Figure 7 but with the power grid included.

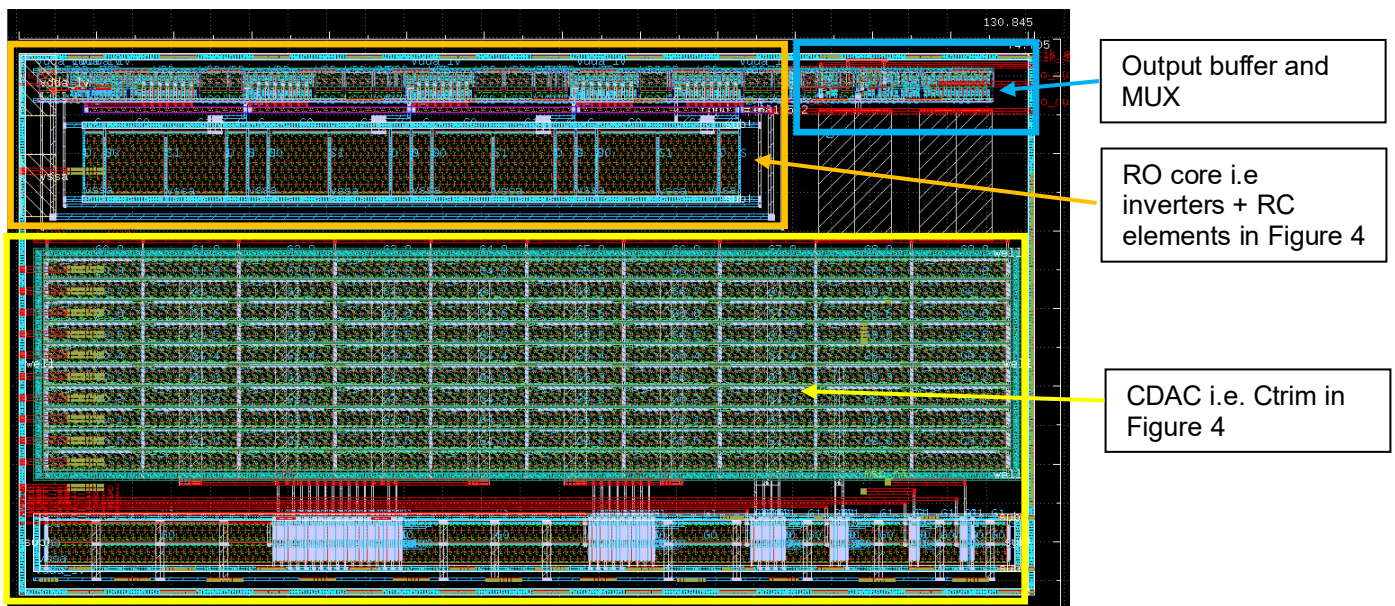


Figure 7 RO top level layout

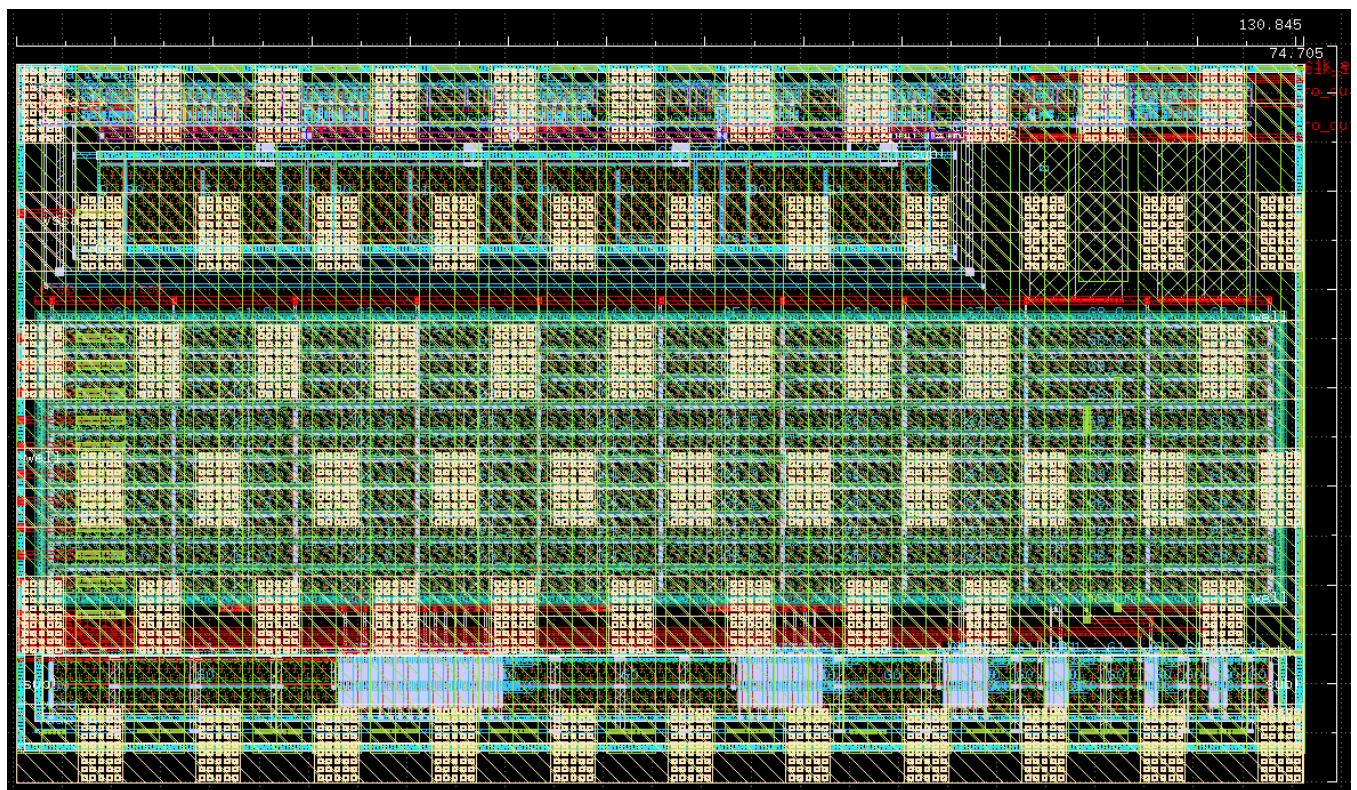


Figure 8 RO top level layout with power grid

6 Testbench simulation results

6.1 Functionality

Figure 9 shows the unbuffered outputs from the block under TT/27C conditions for the mid trim code (dec' 31), pre and post layout. To convert this waveform to a square wave, the unbuffered outputs must pass through a buffer, where Figure 10 shows that buffered output pre and post layout.

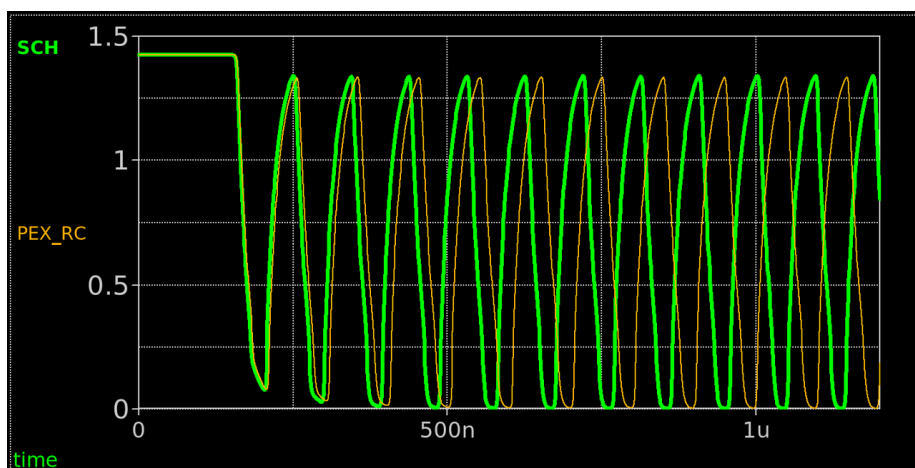


Figure 9 Unbuffered RO output pre (SCH) and post (PEX_RC) layout

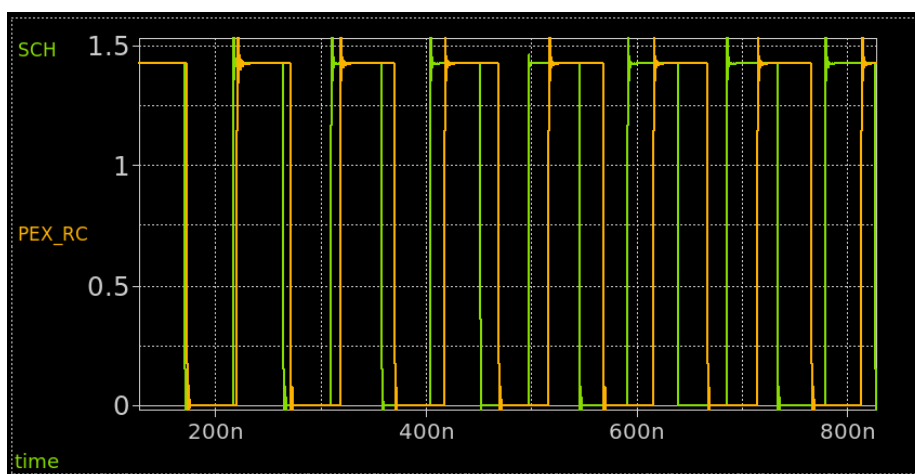


Figure 10 Buffered RO output pre (SCH) and post (PEX_RC) layout

Clearly evident from Figure 10 is a change in frequency pre and post layout. This is quantified in Table 4 which shows fosc to have dropped by 0.5MHz.

Parameter	SCH	PEX	Delta
fosc (MHz)	10.6	10.1	0.5
D (%)	51.56	51.45	0.11
Vptp (V)	1.5	1.5	0
idda (uA)	536	529	7

Table 4 Pre and post layout comparison at TT/27C for mid code ('31)

Root cause of this 0.5MHz reduction is additional parasitic capacitance (C_{par}) of the layout, as shown in (5):

$$f_{osc} = \frac{1}{C_{par} + C_{trim} + (2 * 0.693 * RC * N)} \dots (5)$$

As described in DR2, the impact of post layout parasitics on f_{osc} was anticipated by setting f_{osc} higher than its specified value in schematic (10.6MHz). Table 4 thus confirms this approach by showing f_{osc} to be ~ 10MHz at mid-code, post layout.

Table 4 shows negligible change in duty cycle (D) and output peak to peak (V_{ptp}) as expected, where current consumption (i_{dda}) has decreased by 7uA (again typical of post layout due to parasitic resistance).

6.2 Frequency trim

Trim ability of the RO was next analysed under TT/27C conditions by sweeping the 6b CDAC trim code from 0:2:60 and measuring f_{osc} for each trim code. Resulting tuning curves pre (SCH) and post (RCX) layout are shown in Figure 11.

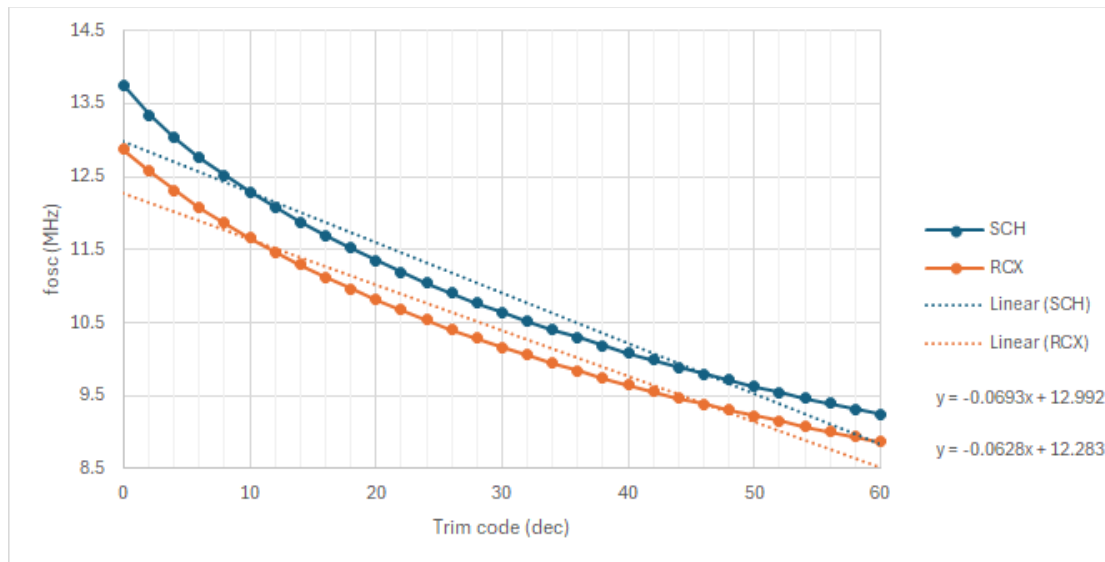


Figure 11 RO tuning curve

First thing to note from Figure 11 is it is less “noisy” looking than the tuning curve presented in DR2. This is due to an improved accuracy setting in ngspice which was used for DR3. Specifically this was achieved by reducing the relative tolerance from its default value of $1e-3$ to $1e-4$ with the below statement:

```
* need reltol of at least 1e-4 to avoid noisy data
.options RELTOL = 1e-4
```

RELTOL sets the relative error tolerance of the program [13] and so reducing it improves accuracy. As this is done at the cost of simulation time, one must be careful not to reduce too much, especially with large netlist typical of those for extracted netlists. When setting the above, one is advised to include “option” inside the .control statement to force ngspice to print out its simulator settings, thus enabling the user to verify the correct tolerance has been set. Assuming the above has been set, the below should be printed to the terminal:

```
Tolerances (relative):
reltol (current) = 0.0001
```

DR3: Ring oscillator

Next is to note that in DR2, codes were swept from 0:1:63 while Figure 11 only sweeps from 0:2:60. Reason for this reduced sweep is due to the long simulation times of the post layout extracted netlist. There is no loss of data here since the purpose of Figure 11 is to show the expected non-linear tuning curve due to the brute force CDAC tuning (as detailed in section 5.2 of [2]).

Using min code values from Figure 11 and simulating the max code, tuning ranges pre and post layout are summarised in Table 5.

freq (MHz)	SCH	RCX
MAX	13.8	12.9
MIN	9.1	8.8
dfreq (+/-%)	20.1	18.9

Table 5 Tuning ranges pre and post layout

Tuning range post layout has reduced by $\pm 1\%$ to $\pm 19\%$. Although this is $\pm 1\%$ out of spec, as shown in Table 6 it is large enough to cover the PVT spread of the RO.

freq (MHz)	SCH	RCX
MAX_31	12.8	12.2
MIN_31	8.7	8.4
dfreq (+/-%)	18.8	18.4

Table 6 RO PVT spread at mid code pre (SCH) and post (RCX) layout

For the RCX max value of Table 6, increasing trim code to '63 reduces the max fosc to 10.6MHz while reducing to '0 increases the min fosc to 10.6MHz. This implies the tuning range needs to be recentred by 600kHz representing a potential improvement to this design.

6.3 Resolution

Due to the non-linear CDAC trim, the LSB is non-linear as summarised in Table 7.

LSB (kHz)	SCH	RCX
MAX	414	298
MIN	71	67
AVE	150	133
Res (%)	1.5	1.3

Table 7 Resolution summary pre (SCH) and post (RCX) layout

Taking the average resolution of 150kHz (SCH) and 133kHz (RCX) shows the 1% spec to be marginally violated.

It should be noted that this number was incorrectly reported as in spec in DR2 as 0.67% and hence in spec. This error was due to the “noisy” data referred to in section 6.2 which was addressed in DR3 by reducing RELTOL as detailed in section 6.2.

6.4 Linearity

Linearity is quantified by looking at the RO's differential non-linearity (DNL) and integral non-linearity (INL) plots, shown in Figure 12 and Figure 13 respectively.

Note that these plots are normalised with respect to the average LSB shown in Table 7.

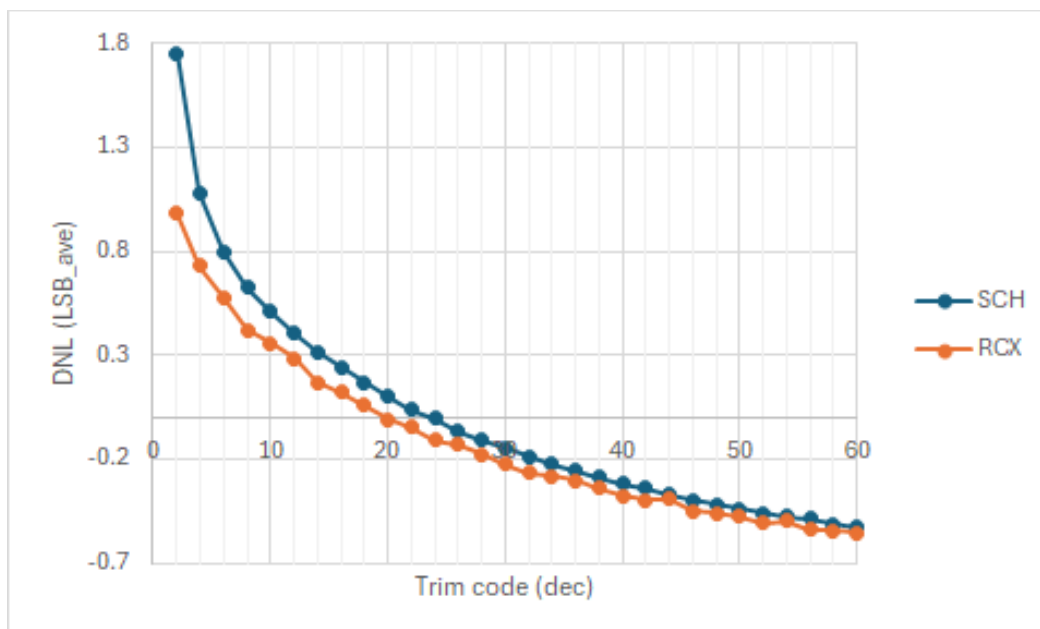


Figure 12 RO DNL pre (SCH) and post (RCX) layout

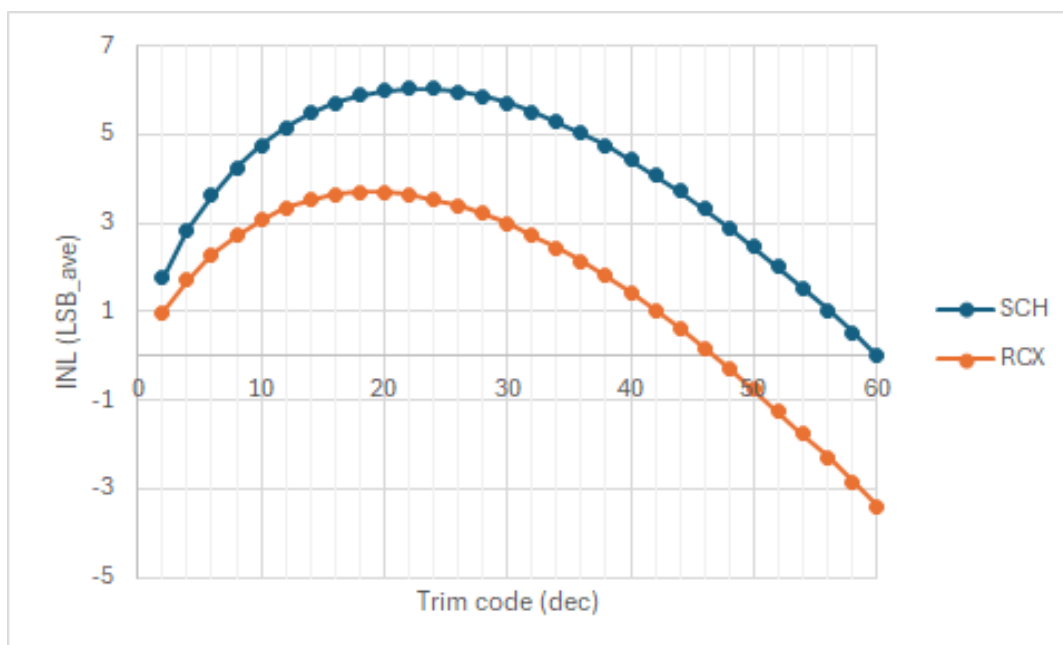


Figure 13 RO INL pre (SCH) and post (RCX) layout

First thing evident from Figure 12 and Figure 13 is they are less “noisy” looking than their corresponding plots presented in DR2. This is due to an improved accuracy setting in ngspice used for DR3, detailed in section 6.2.

The 2nd order non-linear trim achieved by the CDAC is clearly evident from Figure 12 and Figure 13 and summarised in Table 8.

DNL (LSB_ave)	SCH	RCX
MAX	1.8	1.0
MIN	-0.5	-0.6
INL (LSB_ave)	SCH	RCX
MAX	6.0	3.7
MIN	0.0	-3.4

Table 8 Linearity summary pre (SCH) and post (RCX) layout

As shown, INL post layout $\approx \pm 3.5$ LSB_ave which is marginally worse than the ± 3 LSB_ave INL pre-layout. One should note this ± 3 LSB_ave is $\ll$ than the ± 7 LSB_ave quoted in DR2 which was an incorrect number since the measurements were corrupted with simulation accuracy noise, as detailed in section 6.2.

6.5 Frequency spread

As explained in DR2, frequency spread is found by adding the spread in frequency across temperature ($\Delta f_{osc}(T)$) and supply variation ($\Delta f_{osc}(vdda)$) to the trim resolution, or LSB, as per (6):

$$\Delta f_{osc} = \Delta f_{osc}(T) + \Delta f_{osc}(vdda) + LSB_{trim} \dots (6)$$

$\Delta f_{osc}(T)$ from $-40 \rightarrow 125^\circ\text{C}$ is shown in Figure 14 where it can be found to give a spread of $\pm 2.8\%$ post layout ($\pm 2.9\%$ pre layout).

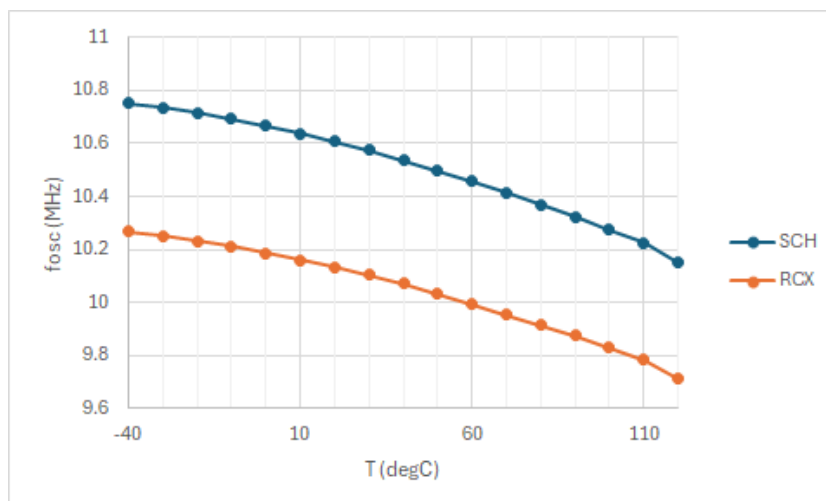


Figure 14 Frequency spread across T pre (SCH) and post (RCX) layout

Again we see Figure 14 to be less “noisy” than its equivalent plot in DR2 due to the improved accuracy of ngspice.

We also see $f_{osc}(T)$ to be reasonably linear, which prompted an investigation into how well it could be used as a temperature sensor. After such analysis, “reasonably” linear was found not to be good enough for an accurate temperature sensor as it produced inaccuracies $> 10^\circ\text{C}$.

$\Delta f_{osc}(vdda)$ is summarized in 9 which shows a supply variation of $\pm 5\%$ to give $\pm 0.23\%$ spread post layout ($\pm 0.27\%$ pre layout). Once again we see the impact of the RC network in the RO core to dramatically reduce the oscillators supply sensitivity (by ~ 4 orders of magnitude) which it shows to be 622Hz/mV post layout (769Hz/mV pre layout).

vdda	SCH	RCX
1.425	10.64	10.15
1.5	10.58	10.11
1.575	10.52	10.06
dfosc (+/-%)	0.27	0.23
dfosc (Hz/mV)	769	622

Table 9 Frequency spread across supply variation

Inserting the above values into (6), and using the resolutions provided in Table 7, gives a frequency spread of:

$$\Delta f_{osc} = 2.8 + 0.23 + 1.3 = \pm 4.33\%$$

which is within the $\pm 5\%$ spec.

Worth noting the pre-layout spread was $\pm 4.67\%$, again in spec.

6.6 Duty cycle and power consumption

The spread in duty cycle (D) and current consumption is summarised in Table 9. This shows negligible impact pre and post layout where all numbers are well within spec.

	MAX		MIN	
	SCH	RCX	SCH	RCX
D (%)	52.2	52.1	51.1	51.04
idda (uA)	621	617	320	323

Table 9 Duty cycle and idda spread pre (SCH) and post (RCX) layout

It should be noted that the idda number was incorrectly quoted in DR2 as being out of spec due to the reduced simulation accuracy which was fixed for DR3.

7 Conclusion

Table 10 summarises the performance of the block at DR3 w.r.t the specifications given in Table 1. Also included for comparison are the DR2 numbers re-simulated for this review, as they are more accurate than those provided in DR2 due to the improved accuracy setting described in 6.2.

This shows all specs to be met apart from trim range and trim resolution. Trim range is marginally less than the $\pm 20\%$ spec ($\pm 18.4\%$) which is not a concern since it covers the $\pm 18.4\%$ intrinsic spread of fosc across PVT. Trim resolution marginally misses the $\pm 1\%$ target, at $\pm 1.3\%$. As such this represents a potential improvement to the design to meet this spec whereby the trim step (i.e. LSB) needs to be reduced.

DR3: Ring oscillator

Design parameter	Unit	Spec	DR2	DR3	Comments
Fosc (typical conditions)	MHz	10	10.6	10.1	
Fosc spread	± %	5	4.67	4.33	Post trim spread
Resolution	%	1	1.5	1.3	Spec violation
Dfosc	± %	20	18.8	18.4	Out of spec but enough to cover fosc intrinsic spread
Duty cycle max	%	55	52.2	52.1	
Duty cycle min	%	45	51.1	51	
idda_hv	uA	1	0.621	0.617	
Power	mW	1.5	0.931	0.925	

Table 10: Design parameters required to meet target specifications

8 References

- [1] <https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf>
- [2] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_2/RO
- [3] <https://ngspice.sourceforge.io/download.html>
- [4] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html
- [5] <https://openvaf.semimod.de/>
- [6] <https://octave.org/download>
- [7] <https://www.klayout.de/build.html>
- [8] <https://opencircuitdesign.com/magic/download.html>
- [9] <https://opencircuitdesign.com/netgen/>
- [10] <https://github.com/IHP-GmbH/ihp-sg13cmos5>
- [11] <https://github.com/IHP-GmbH/IHP-Open-PDK>
- [12] <https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list>
- [13] <https://ngspice.sourceforge.io/docs/ngspice-47-manual.pdf>

9 Revision History

Rev.	Date	Author	Changes, Remarks
A	22.08.2026	Diarmuid Collins	Created

Table 11: Revision history